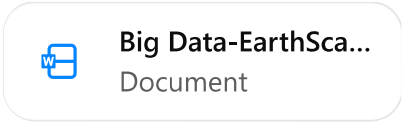
You said:



This is our project for 6th Semester in Aptech
Explain this in easy terms whats exactly we need to do
Core Tech Stack ?
What softwares should we have installed ?
Whats the procedure or working steps ?

ChatGPT said:

You said:

Help me set up the Aptech EarthScape Climate Agency project step by step. OS: Windows 11. RAM: 16 GB. Current installations: VS Code, Python, MongoDB Compass, SQL Server Management Studio. Recommend a compatible setup and start with the first installation step.
Can i use some other tools online instead of installing them like Jupiter Notebook

ChatGPT said:

You said:

```
PS C:\Users\DELL> mkdir "$HOME\Documents\EarthScape-Climate-Agency"
>> cd "$HOME\Documents\EarthScape-Climate-Agency"


    Directory: C:\Users\DELL\Documents


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/16/2026   8:37 PM                EarthScape-Climate-Agency
```

PS C:\Users\DELL\Documents\EarthScape-Climate-Agency> python --version
>> pip --version
Python 3.14.2
pip 25.3 from C:\Python314\Lib\site-packages\pip (python 3.14)
PS C:\Users\DELL\Documents\EarthScape-Climate-Agency>

See terminal and online also shows output:
EarthScape Climate Agency
My online Python environment is working!

ChatGPT said:

You said:

(.venv) PS D:\EarthScape-Climate-Agency> python --version
Python 3.14.2
(.venv) PS D:\EarthScape-Climate-Agency>

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-Service MongoDB -ErrorAction SilentlyContinue

Status Name DisplayName
------ ---- -----------
Running MongoDB MongoDB Server (MongoDB)

PS C:\WINDOWS\system32>

ChatGPT said:

You said:

PS D:\EarthScape-Climate-Agency> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& d:\EarthScape-Climate-Agency\.venv\Scripts\Activate.ps1)
(.venv) PS D:\EarthScape-Climate-Agency> python --version
>> python -m pip --version

```
Python 3.14.2
pip 25.3 from D:\EarthScape-Climate-Agency\.venv\Lib\site-
packages\pip (python 3.14)
(.venv) PS D:\EarthScape-Climate-Agency> python --version
Python 3.14.2
(.venv) PS D:\EarthScape-Climate-Agency> python -c "import sys;
print(sys.executable)"
D:\EarthScape-Climate-Agency\.venv\Scripts\python.exe
(.venv) PS D:\EarthScape-Climate-Agency> python -m pip install flask
pymongo
Collecting flask
  Using cached flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting pymongo
  Downloading pymongo-4.19.0-cp314-cp314-
win_amd64.whl.metadata (10 kB)
Collecting blinker>=1.9.0 (from flask)
  Using cached blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask)
  Downloading click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask)
  Using cached itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9
kB)
Collecting jinja2>=3.1.2 (from flask)
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting markupsafe>=2.1.1 (from flask)
  Downloading markupsafe-3.0.4-cp314-cp314-
win_amd64.whl.metadata (2.8 kB)
Collecting werkzeug>=3.1.0 (from flask)
  Downloading werkzeug-3.1.9-py3-none-any.whl.metadata (4.1 kB)
Collecting dnspython<3.0.0,>=2.7.0 (from pymongo)
  Downloading dnspython-2.9.0-py3-none-any.whl.metadata (5.7 kB)
Using cached flask-3.1.3-py3-none-any.whl (103 kB)
Downloading pymongo-4.19.0-cp314-cp314-win_amd64.whl (831 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 831.1/831.1 kB 19.5
kB/s  0:00:53
Downloading dnspython-2.9.0-py3-none-any.whl (354 kB)
Using cached blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.5.0-py3-none-any.whl (125 kB)
Using cached itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading markupsafe-3.0.4-cp314-cp314-win_amd64.whl (14 kB)
Downloading werkzeug-3.1.9-py3-none-any.whl (228 kB)
```

Installing collected packages: markupsafe, itsdangerous, dnspython, click, blinker, werkzeug, pymongo, jinja2, flask
Successfully installed blinker-1.9.0 click-8.5.0 dnspython-2.9.0 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.4 pymongo-4.19.0 werkzeug-3.1.9

[notice] A new release of pip is available: 25.3 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS D:\EarthScape-Climate-Agency>

ChatGPT said:

You said:

{
  "api": "online",
  "mongodb": "connected"
}

{
  "project": "EarthScape Climate Agency",
  "status": "Backend is running"
}

Both looks fine

Also you are telling we will obtain a small dataset for this project ?? or this is for test

ChatGPT said:

You said:

ok ok

ChatGPT said:

You said:

ok go on

ChatGPT said:

You said:

done

ChatGPT said:

You said:

```
(.venv) PS D:\EarthScape-Climate-Agency> python
.\backend\ingest_climate.py
The CSV file is empty.
(.venv) PS D:\EarthScape-Climate-Agency> python
.\backend\ingest_climate.py
The CSV file is empty.
(.venv) PS D:\EarthScape-Climate-Agency>
```

Its showing this

ChatGPT said:

You said:

CSV rows appear

ChatGPT said:

You said:

```
(.venv) PS D:\EarthScape-Climate-Agency> python -c "from pathlib
import Path; p=Path(r'data/climate_sample.csv').resolve(); print('File:',
p); print('Size:', p.stat().st_size, 'bytes'); print('Content:');
print(repr(p.read_text(encoding='utf-8-sig')))"
File: D:\EarthScape-Climate-Agency\data\climate_sample.csv
Size: 357 bytes
Content:
'\ndate,location,avg_temp_c,rainfall_mm,humidity_pct\n2026-01-
01,Karachi,24.5,0,55\n2026-01-02,Karachi,25.1,0,52\n2026-01-
03,Lahore,16.8,2.5,68\n2026-01-04,Lahore,,5.0,75\n2026-01-
```

```
05,Islamabad,12.4,1.2,80\n2026-01-06,Islamabad,13.1,,77\n2026-01-
07,Karachi,26.0,0,50\n2026-01-08,Lahore,18.2,0,64\n2026-01-
09,Islamabad,14.0,0.8,\n2026-01-10,Karachi,25.6,0,53\n'
(.venv) PS D:\EarthScape-Climate-Agency>
```

ChatGPT said:

You said:

```
(.venv) PS D:\EarthScape-Climate-Agency> python
.\backend\ingest_climate.py

Climate data ingestion complete!
Records processed: 10
Rows skipped: 0
Records currently in MongoDB: 10
(.venv) PS D:\EarthScape-Climate-Agency>
```

ChatGPT said:

You said:

```
from flask import Flask, jsonify
from pymongo import MongoClient
from pymongo.errors import PyMongoError

app = Flask(__name__)

# Connect to the local MongoDB server
client = MongoClient(
    "mongodb://localhost:27017",
    serverSelectionTimeoutMS=3000
)

db = client["earthscape_climate"]
readings = db["climate_readings"]


@app.route("/")
def home():
    return jsonify({
```

```
        "project": "EarthScape Climate Agency",
        "status": "Backend is running"
    })


@app.route("/api/health")
def health():
    try:
        client.admin.command("ping")
        return jsonify({
            "api": "online",
            "mongodb": "connected"
        })
    except PyMongoError:
        return jsonify({
            "api": "online",
            "mongodb": "disconnected"
        }), 503


if __name__ == "__main__":
    app.run(debug=True)
```

This is app.py whats next

ChatGPT said:

You said:


```
{
  "count": 10,
  "data": [
    {
      "avg_temp_c": 25.6,
      "date": "2026-01-10",
      "humidity_pct": 53.0,
      "location": "Karachi",
      "rainfall_mm": 0.0,
      "source": "sample_test_dataset"
    },
    {
```

```
  "avg_temp_c": 14.0,
  "date": "2026-01-09",
  "humidity_pct": null,
  "location": "Islamabad",
  "rainfall_mm": 0.8,
  "source": "sample_test_dataset"
},
{
  "avg_temp_c": 18.2,
  "date": "2026-01-08",
  "humidity_pct": 64.0,
  "location": "Lahore",
  "rainfall_mm": 0.0,
  "source": "sample_test_dataset"
},
{
  "avg_temp_c": 26.0,
  "date": "2026-01-07",
  "humidity_pct": 50.0,
  "location": "Karachi",
  "rainfall_mm": 0.0,
  "source": "sample_test_dataset"
},
{
  "avg_temp_c": 13.1,
  "date": "2026-01-06",
  "humidity_pct": 77.0,
  "location": "Islamabad",
  "rainfall_mm": null,
  "source": "sample_test_dataset"
},
{
  "avg_temp_c": 12.4,
  "date": "2026-01-05",
  "humidity_pct": 80.0,
  "location": "Islamabad",
  "rainfall_mm": 1.2,
  "source": "sample_test_dataset"
},
{
  "avg_temp_c": null,
  "date": "2026-01-04",
```

```
    "humidity_pct": 75.0,
    "location": "Lahore",
    "rainfall_mm": 5.0,
    "source": "sample_test_dataset"
   },
   {
    "avg_temp_c": 16.8,
    "date": "2026-01-03",
    "humidity_pct": 68.0,
    "location": "Lahore",
    "rainfall_mm": 2.5,
    "source": "sample_test_dataset"
   },
   {
    "avg_temp_c": 25.1,
    "date": "2026-01-02",
    "humidity_pct": 52.0,
    "location": "Karachi",
    "rainfall_mm": 0.0,
    "source": "sample_test_dataset"
   },
   {
    "avg_temp_c": 24.5,
    "date": "2026-01-01",
    "humidity_pct": 55.0,
    "location": "Karachi",
    "rainfall_mm": 0.0,
    "source": "sample_test_dataset"
   }
  ]
}
```


ChatGPT said:

You said:

Just a thought we can use React or Python whatever is better for UI presentation

ChatGPT said:

You said:

For EarthScape Climate Agency, use React + Vite with the existing Flask and MongoDB backend. Dashboard style: modern professional dark navy with green accents. React setup: local VS Code using Node.js and Vite. Continue step by step.

ChatGPT said:

You said:

PS D:\EarthScape-Climate-Agency> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& d:\EarthScape-Climate-Agency\.venv\Scripts\Activate.ps1)
(.venv) PS D:\EarthScape-Climate-Agency>

Its showing his

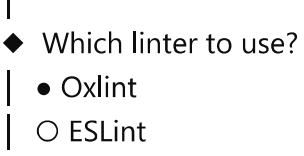
ChatGPT said:

You said:

PS D:\EarthScape-Climate-Agency> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& d:\EarthScape-Climate-Agency\.venv\Scripts\Activate.ps1)
(.venv) PS D:\EarthScape-Climate-Agency> node -v
v24.12.0
(.venv) PS D:\EarthScape-Climate-Agency> npm -v
11.6.2
(.venv) PS D:\EarthScape-Climate-Agency> Get-ChildItem .\frontend
(.venv) PS D:\EarthScape-Climate-Agency>

ChatGPT said:

You said:



```
│  ↑/↓ to navigate • Enter: confirm
└
```

ChatGPT said:

You said:

```
(.venv) PS D:\EarthScape-Climate-Agency> Get-ChildItem .\frontend
(.venv) PS D:\EarthScape-Climate-Agency> npm create vite@latest
frontend -- --template react

> npx
> create-vite frontend --template react

│
◇  Which linter to use?
│  ESLint
│
◇  Install with npm and start now?
│  Yes
│
◇  Scaffolding project in D:\EarthScape-Climate-Agency\frontend...
│
◇  Installing dependencies with npm...

added 143 packages, and audited 144 packages in 28s

31 packages are looking for funding
  run npm fund for details

found 0 vulnerabilities
│
◇  Starting dev server...

> frontend@0.0.0 dev
> vite


  VITE v8.3.4  ready in 294 ms

  ➜  Local:   http://localhost:5173/
```

➜ Network: use --host to expose
➜ press h + enter to show help

ChatGPT said:

You said:

(.venv) PS D:\EarthScape-Climate-Agency> cd D:\EarthScape-Climate-Agency\frontend
>> npm install recharts lucide-react

added 40 packages, and audited 184 packages in 12s

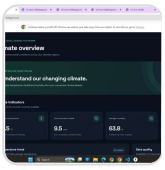
32 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
(.venv) PS D:\EarthScape-Climate-Agency\frontend> npm list recharts lucide-react
frontend@0.0.0 D:\EarthScape-Climate-Agency\frontend
├── lucide-react@1.58.0
└── recharts@3.10.1

(.venv) PS D:\EarthScape-Climate-Agency\frontend>

ChatGPT said:

You said:



Its looking good

ChatGPT said:

You said:

ok ok go on

ChatGPT said:

You said:

Microsoft Windows [Version 10.0.26300.9711]
(c) Microsoft Corporation. All rights reserved.

D:\EarthScape-Climate-Agency>d:\EarthScape-Climate-Agency\.venv\Scripts\activate.bat

(.venv) D:\EarthScape-Climate-Agency>python .\backend\download_climate.py
Downloading historical data for Karachi...

Download failed: NASA POWER returned an unexpected response for Karachi
Check your internet connection and try again.

(.venv) D:\EarthScape-Climate-Agency>python .\backend\download_climate.py
Downloading historical data for Karachi...

Download failed: NASA POWER returned an unexpected response for Karachi
Check your internet connection and try again.

(.venv) D:\EarthScape-Climate-Agency>

ChatGPT said:

You said:

Microsoft Windows [Version 10.0.26300.9711]
(c) Microsoft Corporation. All rights reserved.

D:\EarthScape-Climate-Agency>d:\EarthScape-Climate-Agency\.venv\Scripts\activate.bat

(.venv) D:\EarthScape-Climate-Agency>python .\backend\download_climate.py
Downloading historical data for Karachi...

Download failed: NASA POWER returned an unexpected response for Karachi
Check your internet connection and try again.

(.venv) D:\EarthScape-Climate-Agency>python .\backend\download_climate.py
Downloading historical data for Karachi...

Download failed: NASA POWER returned an unexpected response for Karachi
Check your internet connection and try again.

(.venv) D:\EarthScape-Climate-Agency>python .\backend\download_climate.py
Downloading historical data for Karachi...
NASA POWER response details:


```
{
  "type": "Feature",
  "geometry": {
    "type": "Point",
    "coordinates": [
      67.001,
      24.861,
      53.05
    ]
  },
  "properties": {
    "parameter": {
      "T2M": {
        "20240101": 21.41,
        "20240102": 21.11,
        "20240103": 21.04,
        "20240104": 20.01,
        "20240105": 19.83,
        "20240106": 20.16,
        "20240107": 19.96,
        "20240108": 20.17,
        "20240109": 18.39,
        "20240110": 17.57,
        "20240111": 18.8,
        "20240112": 19.14,
```

"20240113": 19.07,
"20240114": 20.86,
"20240115": 20.69,
"20240116": 21.55,
"20240117": 21.7,
"20240118": 21.66,
"20240119": 21.36,
"20240120": 21.82,
"20240121": 21.54,
"20240122": 21.21,
"20240123": 20.96,
"20240124": 20.02,
"20240125": 19.84,
"20240126": 20.59,
"20240127": 21.21,
"20240128": 20.93,
"20240129": 22.77,
"20240130": 22.14,
"20240131": 22.28,
"20240201": 22.57,
"20240202": 21.91,
"20240203": 22.38,
"20240204": 21.87,
"20240205": 20.85,
"20240206": 21.64,
"20240207": 21.42,
"20240208": 20.25,
"20240209": 20.17,
"20240210": 21.28,
"20240211": 21.51,
"20240212": 21.61,
"20240213": 21.76,
"20240214": 21.7,
"20240215": 21.63,
"20240216": 22.35,
"20240217": 21.86,
"20240218": 22.75,
"20240219": 23.33,
"20240220": 22.09,
"20240221": 21.87,
"20240222": 21.11,
"20240223": 20.58,

"20240224": 21.52,
"20240225": 20.6,
"20240226": 20.88,
"20240227": 20.61,
"20240228": 22.87,
"20240229": 22.81,
"20240301": 20.57,
"20240302": 16.43,
"20240303": 16.25,
"20240304": 18.04,
"20240305": 20.23,
"20240306": 20.17,
"20240307": 21.73,
"20240308": 22.16,
"20240309": 22.85,
"20240310": 22.25,
"20240311": 22.81,
"20240312": 23.45,
"20240313": 22.2,
"20240314": 22.78,
"20240315": 23.45,
"20240316": 23.9,
"20240317": 25.3,
"20240318": 25.45,
"20240319": 25.49,
"20240320": 25.4,
"20240321": 25.5,
"20240322": 25.4,
"20240323": 26.11,
"20240324": 26.78,
"20240325": 27.44,
"20240326": 27.3,
"20240327": 26.46,
"20240328": 26.53,
"20240329": 24.93,
"20240330": 25.33,
"20240331": 25.54,
"20240401": 26.09,
"20240402": 26.11,
"20240403": 25.84,
"20240404": 27.07,
"20240405": 26.77,

"20240406": 26.49,
"20240407": 26.83,
"20240408": 27.51,
"20240409": 27.73,
"20240410": 27.15,
"20240411": 27.4,
"20240412": 27.23,
"20240413": 27.05,
"20240414": 26.13,
"20240415": 25.9,
"20240416": 27.74,
"20240417": 27.98,
"20240418": 26.91,
"20240419": 26.36,
"20240420": 26.99,
"20240421": 27.88,
"20240422": 27.49,
"20240423": 27.85,
"20240424": 27.64,
"20240425": 27.04,
"20240426": 27.65,
"20240427": 27.98,
"20240428": 29.24,
"20240429": 28.97,
"20240430": 29.49,
"20240501": 28.67,
"20240502": 28.92,
"20240503": 28.09,
"20240504": 28.71,
"20240505": 28.82,
"20240506": 29.81,
"20240507": 30.07,
"20240508": 29.98,
"20240509": 30.05,
"20240510": 29.22,
"20240511": 29.53,
"20240512": 29.97,
"20240513": 29.99,
"20240514": 29.62,
"20240515": 30.21,
"20240516": 30.59,
"20240517": 31.1,

"20240518": 30.89,
"20240519": 29.99,
"20240520": 30.73,
"20240521": 32.

Download failed: NASA POWER returned an unexpected response for Karachi
Check your internet connection and try again.

(.venv) D:\EarthScape-Climate-Agency>

ChatGPT said:

You said:

(.venv) D:\EarthScape-Climate-Agency>python .\backend\download_climate.py
Downloading historical data for Karachi...
Downloading historical data for Lahore...
Downloading historical data for Islamabad...

Download complete.
Cities: 3
Records downloaded: 1098
Output file: D:\EarthScape-Climate-Agency\data\climate_historical.csv
Source: NASA POWER

(.venv) D:\EarthScape-Climate-Agency>

ChatGPT said:

You said:

(.venv) D:\EarthScape-Climate-Agency>powershell -Command "Import-Csv .\data\climate_historical.csv | Select-Object -First 5 | Format-Table"

date location avg_temp_c min_temp_c max_temp_c rainfall_mm humidity_pct source
---- -------- ---------- ---------- ---------- ---------- ------------ ------

```
2024-01-01 Karachi  21.41      16.95      26.88      0.0        42.71
NASA POWER
2024-01-02 Karachi  21.11      17.18      25.33      0.02       65.67
NASA POWER
2024-01-03 Karachi  21.04      16.92      25.71      0.01       63.21
NASA POWER
2024-01-04 Karachi  20.01      16.06      24.83      0.01       41.51
NASA POWER
2024-01-05 Karachi  19.83      15.3       24.17      0.08       43.01
NASA POWER
```

```
(.venv) D:\EarthScape-Climate-Agency>powershell -Command
"Import-Csv .\data\climate_historical.csv | Group-Object location |
Select-Object Name,Count | Format-Table"

Name      Count
----      -----
Karachi     366
Lahore      366
Islamabad   366
```

```
(.venv) D:\EarthScape-Climate-Agency>
```

ChatGPT said:

You said:

```
(.venv) D:\EarthScape-Climate-Agency>python
.\backend\ingest_climate.py
Traceback (most recent call last):
  File "D:\EarthScape-Climate-Agency\backend\ingest_climate.py", line
5, in <module>
    from pymongo import MongoClient
ModuleNotFoundError: No module named 'pymongo'

(.venv) D:\EarthScape-Climate-Agency>
```

```
import csv
```

```
from datetime import date
from pathlib import Path

from pymongo import MongoClient

# Find the project folder automatically
PROJECT_DIR = Path(__file__).resolve().parent.parent
CSV_FILE = PROJECT_DIR / "data" / "climate_historical.csv"

# Connect to MongoDB
client = MongoClient(
    "mongodb://localhost:27017",
    serverSelectionTimeoutMS=3000
)

db = client["earthscape_climate"]
readings = db["climate_readings"]

REQUIRED_COLUMNS = {
    "date",
    "location",
    "avg_temp_c",
    "rainfall_mm",
    "humidity_pct",
}


def parse_number(value, column_name, row_number):
    """Convert a CSV value to a number, or None if it is empty."""
    value = value.strip()

    if value == "":
        return None

    try:
        number = float(value)
    except ValueError:
        raise ValueError(
            f"Invalid number in '{column_name}' on CSV row
{row_number}"
        )
```

```
    if column_name == "humidity_pct" and not 0 <= number <= 100:
        raise ValueError(
            f"Humidity must be between 0 and 100 on row {row_number}"
        )

    if column_name == "rainfall_mm" and number < 0:
        raise ValueError(
            f"Rainfall cannot be negative on row {row_number}"
        )

    return number


def main():
    if not CSV_FILE.exists():
        print(f"CSV file not found: {CSV_FILE}")
        return

    # Check the MongoDB connection
    client.admin.command("ping")

    # Prevent duplicate records for the same date and location
    readings.create_index(
        [("date", 1), ("location", 1)],
        unique=True
    )

    processed = 0
    skipped = 0

    with CSV_FILE.open("r", encoding="utf-8-sig", newline="") as file:
        reader = csv.DictReader(file)

        if not reader.fieldnames:
            print("The CSV file is empty.")
            return

        missing_columns = REQUIRED_COLUMNS -
set(reader.fieldnames)

        if missing_columns:
            print("Missing CSV columns:", ",
```

```
".join(sorted(missing_columns)))
            return

        for row_number, row in enumerate(reader, start=2):
            try:
                record_date = date.fromisoformat(row["date"].strip())
                location = row["location"].strip()

                if not location:
                    raise ValueError("Location cannot be empty")

                record = {
                    "date": record_date.isoformat(),
                    "location": location,
                    "avg_temp_c": parse_number(
                        row["avg_temp_c"], "avg_temp_c", row_number
                    ),
                    "rainfall_mm": parse_number(
                        row["rainfall_mm"], "rainfall_mm", row_number
                    ),
                    "humidity_pct": parse_number(
                        row["humidity_pct"], "humidity_pct", row_number
                    ),
                    "source": "sample_test_dataset",
                }

                readings.replace_one(
                    {
                        "date": record["date"],
                        "location": record["location"],
                    },
                    record,
                    upsert=True,
                )

                processed += 1

            except (ValueError, TypeError) as error:
                skipped += 1
                print(f"Skipping CSV row {row_number}: {error}")

    print("\nClimate data ingestion complete!")
```

```
    print(f"Records processed: {processed}")
    print(f"Rows skipped: {skipped}")
    print(f"Records currently in MongoDB:
{readings.count_documents({})}")


if __name__ == "__main__":
    try:
        main()
    except Exception as error:
        print(f"Error: {error}")
    finally:
        client.close()
```

ChatGPT said:

You said:

```
(.venv) D:\EarthScape-Climate-Agency>python
.\backend\ingest_climate.py

Climate data ingestion complete!
Records processed: 1098
Rows skipped: 0
Records currently in MongoDB: 1108

(.venv) D:\EarthScape-Climate-Agency>
```

ChatGPT said:

You said:

```
{
  "api": "online",
  "mongodb": "connected"
}
```

ChatGPT said:

You said:



Pasted markdown....
Document

Also see

ChatGPT said:

You said:

```
from flask import Flask, jsonify, request
from pymongo import MongoClient
from pymongo.errors import PyMongoError

app = Flask(__name__)

# Connect to MongoDB
client = MongoClient(
    "mongodb://localhost:27017",
    serverSelectionTimeoutMS=3000
)

db = client["earthscape_climate"]
readings = db["climate_readings"]


@app.route("/")
def home():
    return jsonify({
        "project": "EarthScape Climate Agency",
        "status": "Backend is running"
    })


@app.route("/api/health")
def health():
    try:
```

```
        client.admin.command("ping")
        return jsonify({
            "api": "online",
            "mongodb": "connected"
        })
    except PyMongoError:
        return jsonify({
            "api": "online",
            "mongodb": "disconnected"
        }), 503


@app.route("/api/climate", methods=["GET"])
def get_climate_data():
    try:
        # Optional filters
        location = request.args.get("location")
        limit_text = request.args.get("limit", "100")

        try:
            limit = int(limit_text)
        except ValueError:
            return jsonify({
                "error": "limit must be a whole number"
            }), 400

        if not 1 <= limit <= 500:
            return jsonify({
                "error": "limit must be between 1 and 500"
            }), 400

        query = {}

        if location:
            query["location"] = {
                "$regex": f"^{location}$",
                "$options": "i"
            }

        # Retrieve newest dates first
        cursor = readings.find(
            query,
```

```
            {"_id": 0}
        ).sort("date", -1).limit(limit)

        records = list(cursor)

        return jsonify({
            "count": len(records),
            "data": records
        })

    except PyMongoError:
        return jsonify({
            "error": "Unable to retrieve climate data"
        }), 503


if __name__ == "__main__":
    app.run(debug=True)
```

ChatGPT said:

You said:

"count": 1098,

ChatGPT said:

You said:

Pasted text.txt
Document

Its lookin good

ChatGPT said:

You said:

Big Data-EarthSca...
Document

ok ok whatever meets under the requiement file we will be more than happy to do it
Chk again for better context

ChatGPT said:

You said:

Ok ok start

ChatGPT said:

You said:

PS D:\EarthScape-Climate-Agency> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& d:\EarthScape-Climate-Agency\.venv\Scripts\Activate.ps1)
(.venv) PS D:\EarthScape-Climate-Agency> python .\backend\audit_climate_data.py

=== EARTHSCAPE DATA QUALITY AUDIT ===
Total records: 1108

Records by source:
  sample_test_dataset: 10
  NASA POWER: 1098

Records by location:
  Karachi: 370
  Lahore: 369
  Islamabad: 369

Earliest date: 2024-01-01
Latest date: 2026-01-10

Missing values by field:
  avg_temp_c: 1
  min_temp_c: 1108
  max_temp_c: 1108
  rainfall_mm: 1
  humidity_pct: 1

```
Audit completed. No records were changed.
(.venv) PS D:\EarthScape-Climate-Agency>
```

ChatGPT said:

You said:

```
(.venv) PS D:\EarthScape-Climate-Agency> python
.\backend\inspect_climate_records.py

=== NASA POWER RECORD ===
{'avg_temp_c': 11.83,
 'date': '2024-01-01',
 'humidity_pct': 54.71,
 'location': 'Islamabad',
 'rainfall_mm': 0.0,
 'source': 'NASA POWER'}

=== TEST DATA RECORD ===
{'avg_temp_c': 24.5,
 'date': '2026-01-01',
 'humidity_pct': 55.0,
 'location': 'Karachi',
 'rainfall_mm': 0.0,
 'source': 'sample_test_dataset'}
(.venv) PS D:\EarthScape-Climate-Agency>
```

ChatGPT said:

You said:

```
import csv
from datetime import date
from pathlib import Path

from pymongo import MongoClient

# Find the project folder automatically
PROJECT_DIR = Path(__file__).resolve().parent.parent
CSV_FILE = PROJECT_DIR / "data" / "climate_historical.csv"
```

```
# Connect to MongoDB
client = MongoClient(
    "mongodb://localhost:27017",
    serverSelectionTimeoutMS=3000
)

db = client["earthscape_climate"]
readings = db["climate_readings"]

REQUIRED_COLUMNS = {
    "date",
    "location",
    "avg_temp_c",
    "rainfall_mm",
    "humidity_pct",
}


def parse_number(value, column_name, row_number):
    """Convert a CSV value to a number, or None if it is empty."""
    value = value.strip()

    if value == "":
        return None

    try:
        number = float(value)
    except ValueError:
        raise ValueError(
            f"Invalid number in '{column_name}' on CSV row
{row_number}"
        )

    if column_name == "humidity_pct" and not 0 <= number <= 100:
        raise ValueError(
            f"Humidity must be between 0 and 100 on row {row_number}"
        )

    if column_name == "rainfall_mm" and number < 0:
        raise ValueError(
            f"Rainfall cannot be negative on row {row_number}"
```

```
        )

    return number


def main():
    if not CSV_FILE.exists():
        print(f"CSV file not found: {CSV_FILE}")
        return

    # Check the MongoDB connection
    client.admin.command("ping")

    # Prevent duplicate records for the same date and location
    readings.create_index(
        [("date", 1), ("location", 1)],
        unique=True
    )

    processed = 0
    skipped = 0

    with CSV_FILE.open("r", encoding="utf-8-sig", newline="") as file:
        reader = csv.DictReader(file)

        if not reader.fieldnames:
            print("The CSV file is empty.")
            return

        missing_columns = REQUIRED_COLUMNS -
set(reader.fieldnames)

        if missing_columns:
            print("Missing CSV columns:", ",
".join(sorted(missing_columns)))
            return

        for row_number, row in enumerate(reader, start=2):
            try:
                record_date = date.fromisoformat(row["date"].strip())
                location = row["location"].strip()
```

```
                if not location:
                    raise ValueError("Location cannot be empty")

                record = {
                    "date": record_date.isoformat(),
                    "location": location,
                    "avg_temp_c": parse_number(
                        row["avg_temp_c"], "avg_temp_c", row_number
                    ),
                    "rainfall_mm": parse_number(
                        row["rainfall_mm"], "rainfall_mm", row_number
                    ),
                    "humidity_pct": parse_number(
                        row["humidity_pct"], "humidity_pct", row_number
                    ),
                    "source": row.get("source", "").strip() or "NASA POWER",
                }

                readings.replace_one(
                    {
                        "date": record["date"],
                        "location": record["location"],
                    },
                    record,
                    upsert=True,
                )

                processed += 1

            except (ValueError, TypeError) as error:
                skipped += 1
                print(f"Skipping CSV row {row_number}: {error}")

    print("\nClimate data ingestion complete!")
    print(f"Records processed: {processed}")
    print(f"Rows skipped: {skipped}")
    print(f"Records currently in MongoDB:
{readings.count_documents({})}")


if __name__ == "__main__":
    try:
```

```
    main()
except Exception as error:
    print(f"Error: {error}")
finally:
    client.close()
```

ChatGPT said:

You said:

```
(.venv) PS D:\EarthScape-Climate-Agency> python
.\backend\ingest_climate.py

Climate data ingestion complete!
Records processed: 1098
Rows skipped: 0
Records currently in MongoDB: 1108
(.venv) PS D:\EarthScape-Climate-Agency> python
.\backend\inspect_climate_records.py

=== NASA POWER RECORD ===
{'avg_temp_c': 11.83,
 'date': '2024-01-01',
 'humidity_pct': 54.71,
 'location': 'Islamabad',
 'max_temp_c': 19.23,
 'min_temp_c': 8.16,
 'rainfall_mm': 0.0,
 'source': 'NASA POWER'}

=== TEST DATA RECORD ===
{'avg_temp_c': 24.5,
 'date': '2026-01-01',
 'humidity_pct': 55.0,
 'location': 'Karachi',
 'rainfall_mm': 0.0,
 'source': 'sample_test_dataset'}
(.venv) PS D:\EarthScape-Climate-Agency>
```

ChatGPT said:

You said:

```
(.venv) PS D:\EarthScape-Climate-Agency> python
.\backend\data_quality_report.py

=== EARTHSCAPE DATA QUALITY REPORT ===

Source: NASA POWER
Total records: 1098
  avg_temp_c: 0 missing (0.00%)
  min_temp_c: 0 missing (0.00%)
  max_temp_c: 0 missing (0.00%)
  rainfall_mm: 0 missing (0.00%)
  humidity_pct: 0 missing (0.00%)
Locations: {'Karachi': 366, 'Lahore': 366, 'Islamabad': 366}

Source: sample_test_dataset
Total records: 10
  avg_temp_c: 1 missing (10.00%)
  min_temp_c: 10 missing (100.00%)
  max_temp_c: 10 missing (100.00%)
  rainfall_mm: 1 missing (10.00%)
  humidity_pct: 1 missing (10.00%)
Locations: {'Karachi': 4, 'Lahore': 3, 'Islamabad': 3}

Report complete. No records were changed.
(.venv) PS D:\EarthScape-Climate-Agency>
```

ChatGPT said: